

Homework

EVOLVING DOCUMENT

Plaksha: Robot motion simulation and optimization

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[Hyperlinks look like this.](#)

1 Homework Standards and Format

The output of your work this semester is a portfolio of problem solutions which you understand. All homework is aimed at that. Each problem solution should be in a folder. The name of the folder should be, for example: “2. Numerical Derivative”. Inside the folder should be any relevant Matlab files and a document. The document could be a pdf, Word, Google Docs, LaTeX w/ pdf, or a scan of hand written work. Anything that is clear. Figures could be separate files that you refer to in the text or could be included in the text file. The general goal is that in 10 years you can open a folder and understand the problem and the solution.

Solution quality. You need a brief problem restatement so that a reader knows what you are trying to do. An ideal solution will convince someone who has no reason to trust you, that your solution is correct. It will be such that someone who doesn’t know how to do the problem can learn how from your solution. It will explore the problem from various angles, considering special and extreme cases, general trends etc. You need a brief problem restatement. You need any graphs that illustrate important features of the solution as well as a still shot of any simulation. And you need enough words to be clear.

Code. Ideally, your code should be so well commented that anyone who does not know how to write the code, can learn it from your code and comments.

Lets get real. If I, Andy, was to do the best job I can do on the problems below — problems which I know how to solve — using the standards above, it would take me 4-8 hours for most of the problem. It takes a lot of time to be clear, complete and exploratory. That is, I know that you do not have time to do a perfect job on all of the problems. What is written above is a standard you want to vaguely aim for. It is not something that you can do regularly and stay inside the time you should give to this course.

2 Problems

1. **Matlab.** Master all parts of the Pratap Matlab book that we posted on Moodle
2. **Numerical derivative.** Use the definition of derivative to calculate the derivative of $\sin(x)$ at $x = 1$.
 - (a) Compare that to the exact value (exact value is $\cos(1)$). [Practically speaking, ‘exact’ means the full computer precision before the compounding of roundoff errors. That is, it is the roundoff error of numbers and of elementary functions.]
 - (b) Plot error vs h (log log plot) and note that there is a best h because of the competition between Method error and roundoff error.
 - (c) Challenge: Using the Matlab resolution of Matlab’s smallest number “eps” (look it up), estimate the best h for this calculation and compare It with the minimum on your graph.
3. **Tutorial problem.** Complete the exercises from the Tutorial on Thursday morning 29 January.
4. **Euler solver.** Demonstrate that you can write code equivalent to the code shared with you on 30 January. That code separates the solution of an ODE into
 - i) a script (problem set up) and two matlab functions:
 - ii) The Right Hand Side (RHS) function; and
 - iii) The Euler solver (that can solve any set of first order ODEs). If you do that first, you can use it to make the error plots above. You should be able to write this using Matlab help and no more (no Google, no GPT, no copilot).
5. **Make up a problem.** Cook up, or find, some ODE, or set of first order ODEs, and solve them and explore something hopefully-interesting about the solution.

6. Higher order methods.

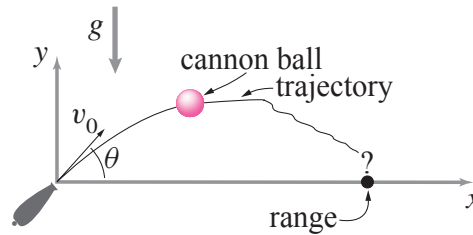
- (a) Calculate the derivative of $\sin(x)$ using central difference. Make a log log plot of error vs h . On one plot show Euler's method and the central difference method. You should notice that
- the best h is slightly bigger with central difference than Euler
 - for given h the error is much smaller with central difference
 - the slope of the curve for central difference is about 2, twice the slope of the curve for Euler (about 1). This indicates that the error for Euler is proportional to h , and for central difference the error is proportional to h^2 . Hence Central Difference is a so called 'second order' method, whereas with Euler, the slope is one indicating that error is proportional to h so it is called a 'first order' method
- (b) Write a new Matlab function that solves arbitrary ODEs using 'the midpoint method'. This is only a slight modification of the Euler solver. Test it with the differential equation $\dot{x} = x$ with initial condition $x(0) = 1$, evaluated at $t = 1$. On one log log plot show the error as a function of h for both the Euler method and the midpoint method.

7. **Spring and mass (2D).** One end of a negligible-mass spring (k, L_0) is pinned to the origin, the other to a mass (m). There is gravity (g). Initial Conditions (ICs): The initial position is $\vec{r}_0 = x_0\hat{i} + y_0\hat{j}$, and the initial velocity is $\vec{v}_0 = v_{x0}\hat{i} + v_{y0}\hat{j}$. Motion starts at $t = 0$ and ends at t_{end} .

- (a) Find the Equations of Motion (EoM);
- (b) Assume all parameters and IC's above are given.
- Plot the trajectory of the mass.
 - Animate the trajectory of the mass.
- (c) How many ways can you think of checking the numerical solution? Find as many as you can, and do the check. A list of things to check is started here:
- $k = 0$, all else arbitrary: The motion is parabolic flight (including falling straight down as a special case) [Why? The system is then just ballistics from freshman physics];
 - $x_0 = 0, v_{x0} = 0$, all else arbitrary: The motion stays on the y axis [Why? There is no force in the x direction if the mass is on the y axis. Because the initial velocity has no x component, the mass never leaves the y axis];
 - $g = 0, \vec{v}_0 = \vec{0}$, all else is arbitrary: The motion stays on a radial line. And, if the motion does not cross the origin, the motion is that of a harmonic oscillator (sinusoidal oscillations, check by plotting, say x vs t . [Why? (Advanced) Write the EoM and EoMs in polar coordinates
 $\Rightarrow m\ddot{r} = -k(r - L_0)$
 $\Rightarrow$ the harmonic oscillator equation, $m\ddot{r}^* = -kr^*$, where $r^* \equiv r - L_0$].
 - $L_0 = 0$, all else is arbitrary: ____? _____. [Why? ____? _____.] The question marks indicate the interesting features of the solution and the explanation for those interesting features.
 - etc.*

- vi. *etc.*
- vii. ...

8. **Cannon ball.** A cannon ball m is launched at angle θ and speed v_0 . It is acted on by gravity g and a viscous drag with magnitude $|cv|$.
- (a) Find position vs time analytically (optional, for those who like theory)
 - (b) Find a numerical solution using $\theta = \pi/4$, $v_0 = 1$ m/s, $g = 1$ m/s², $m = 1$ kg, $c = 1$ kg/s.
 - (c) Compare the numeric and analytic solutions. At $t = 2$ how big is the error? How does the error depend on specified tolerances or step sizes?
 - (d) Use larger and larger values of v_0 and for each trajectory choose a time interval so the canon at least gets back to the ground. Plot the trajectories (using equal scale for the x and y axis. Plot all curves on one plot. As $v \rightarrow \infty$ what is the eventual shape? [Hint: the answer is simple and interesting.]
 - (e) For any given v_0 there is a best launch angle θ^* for maximizing the range. As $v_0 \rightarrow \infty$ to what angle does θ^* tend? Justify your answer as best you can with careful numerics, analytical work, or both.



9. **Statics vs Dynamics.** 2D. Equilibrium and dynamics of a point mass m at $\vec{r}_C$. Gravity $-g\hat{j}$ points down. Points A and B are anchored at $\vec{r}_A$ and $\vec{r}_B$. Springs and parallel dashpots connect the mass to A and B. The spring constants are k_A and k_B . Their rest lengths are L_A and L_B . Parallel to the springs are dashpots c_A and c_B . The mass also has a drag force proportional to speed through the air c_d .

The dynamic state of the system is described with $z = [x, y, v_x, v_y]'$ and the static state by $z = [x, y, 0, 0]'$.

- (a) **Basic Statics.** Given parameters and an initial guess, find an equilibrium position of this system. If you have an example you like, post it and people can compare solutions.
- (b) **Basic Dynamics.** Given parameters and an initial condition, find the motion.
- (c) **All equilibria.** Given parameters, attempt to find all equilibria. How many are there? By varying parameters, what is the most and least number of equilibrium points there can be?
- (d) **Stability of equilibrium.** Given an equilibrium point, find it's stability by using a dynamic simulation that starts near the equilibrium; (former part (b) deleted and moved to a later problem.).

Make appropriate plots, animations and comparisons.

10. **Linear ODEs.** Although we are emphasizing numerical solution of arbitrarily complex ordinary differential equations, there are some simple equations you should know the solution to. Most of these are summarized on pages 198 - 201 of the Ruina/Pratap book. The method of solution that I recommend is to guess, then constrain your guess by plugging it into the ODE and finding any free constants. Find the general solution to these ODEs. You should *know* these, any time and any place, for the rest of your technical life. The first problem below shows the idea. So does one of the tutorials you have done.

- (a) What is the general solution to the ODE $\dot{x} = 7x$?

Solution. Guess: $x = Ce^{\lambda t}$ (where C and λ are as yet undetermined constants). Plug guess into the ODE:

$$\begin{aligned} d/dt(Ce^{\lambda t}) &\stackrel{?}{=} 7Ce^{\lambda t} \\ \Rightarrow C\lambda e^{\lambda t} &\stackrel{?}{=} 7Ce^{\lambda t} \\ \Rightarrow \lambda &= 7 \end{aligned}$$

This implies that $x = Ce^{\lambda t}$ solves the ODE $\dot{x} = 7x$ for $\lambda = 7$ and for *any* value of C .

- (b) Find a solution for $\dot{x} = -3x$. Find the general solution. Use that to solve the Initial Value Problem (IVP): $x(0) = 5$. Compare the solution with ode45 and plot both on the same plot (one of them should be a dotted line because they may be so similar that it would/will look like a single solution)
- (c) Consider the ODE: $\ddot{x} = 9x$. Note, for this problem and the remaining problems: For these linear homogeneous equations, any linear combination of two solutions is also a solution. Find the general solution. Use guessing, don't try to derive the solution. How to guess? Think of a function, put it in the equation, and see if it 'satisfies' the equation. If not, try another. You should find two independent solutions (one is not a multiple of the other). Note, that you can also write the general solution using $\cosh(t)$ and $\sinh(t)$ (Why?).
- (d) Same as above for $\ddot{x} = -4x$. Note, unlike the case with the first order ODE, the sign change changes the whole structure of the solution.
- (e) Nowhere in this and the following subproblems should there be any use of determinants or polynomials. Only guessing or other methods. Find a solution to $\dot{z} = Az$, where z is a two element vector and A is a given 2×2 matrix of constants.
[Hint: guess that $z = e^{\lambda t} \cdot v$, where v is a yet to be determined two element constant vector.]. At this point leave this abstract.
- (f) Still abstract, find the general solution.
- (g) Still abstract, solve the initial value problem $z(t = 0) = z_0$ where z_0 is a given column vector of two numbers.
- (h) Repeat the whole sequence above with $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ and $z_0 = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$.

- (i) Compare the above solution with `ode45`, either with time plots or with phase plane trajectories, plotting the exact solution on top of the numerical solution, making one of the two with a dotted line.
- (j) Consider this IVP:

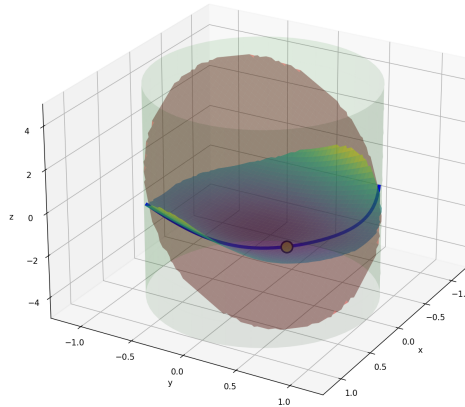
$$\dot{z} = Az \quad \text{with} \quad z(0) = z_0.$$

Make up a random 4×4 matrix A . Make up a random initial condition z_0 . Solve the IVP two ways: 1) with eigen-values and eigenvectors (using the MATLAB command `eig`) and exponential functions; and with `ode45`. Compare the solutions on the same plot (plot whatever you like), making one of the curves dotted.

11. Linear stability of springs and mass. Back to the situation of problem 9.

- (a) Pick parameters of your choice (with at least one of the two springs having non-zero rest length or at least one dashpot being non-zero).
- (b) Mark both anchor points. Draw lines from the anchors to the mass representing the springs. Make the mass a clear bright disk. Draw thin-line circles around both anchor points representing the rest lengths of the springs. Mark all fixed points that you find. Draw the path of the mass from the start up to the present time. Make the animation as big, smooth and clear as you can. Gloat with pride and show a friend or relative.
- (c) Try various initial conditions and get a sense of where the equilibrium points are in relation to the circles, and how the motion behaves in relation to the anchor points and the equilibrium points.
- (d) Bonus. Sometimes there are equilibrium points *very* close to anchor points. Can you explain why?
- (e) Using the Jacobian supplied by `fsolve` at one of the stable fixed points, solve the linearized equations for motion for an initial condition near the fixed point.
- (f) Using the same initial condition, use `ODE45` to find the motion.
- (g) Find an unstable fixed point. Repeat the exercise above using a long enough time to see some complexity in the motion.
- (h) Setting both spring rest-lengths to zero and both dashpots to zero, linearize about the stable fixed point. Solve the same IVP for some initial condition far from the fixed point (and with non zero initial velocity). Compare with the solution of the same IVP using `ODE45`. Plot the two solutions on the same plot, with one curve dotted.
- (i) Bonus. If both springs have zero rest length, and the two dashpots are zero, the whole system turns out to be linear. Why?

12. Numerical optimization. Consider the surface $z = (x(x-y))^2 - \exp(-x^2 - y^2) - \cos(x)$. Consider the plane $x + y + z/3 = 0$. Find the minimum value of z that is on the surface, on the plane and inside the unit circle. Find the x and y (called 'argmin'). Do this with a grid search (a bit tricky so will take some thought). Then learn `fmincon` and try again. Start with easier problems for practice. If you have time within the 40 hours you should have spent on this course so far, make a 3D plot showing the surface, the plane and the minimum point. According to Claude the answer is $(x, y, z) \approx (0.236, 0.364, -1.800)$.



13. **Periodic Solutions.** Finding periodic motions. In all cases, try to be clear in your mind, and in what you show, what things are due to numerical approximation and what are real phenomena. For example, If something is actually periodic, but appears as only approximately periodic, it will become closer and closer to periodic if you increase your integration accuracy.

Note: Unless you know a shortcut, the only way to find a periodic solution is by root finding of some kind.

The actual homework problem here, is the last item. What is above that is necessary preparation.

- Verify, using numerical solutions, that for $\ddot{\vec{r}} = -(k/m)\vec{r}$ that for any and all initial conditions you get periodic motions (straight lines, circles, or ellipses). Extra credit, verify this with pencil and paper. This is a 'linear' central force'.
- Consider $\ddot{\vec{r}} = -G(M/r^3)\vec{r}$ (this is an inverse square central gravitational force towards a fixed mass M . The mass m_1 of the moving mass drops out because the gravitational force is proportional to mass *Looks like inverse cube, why called 'inverse square'?*). Verify with numerical solutions that for any and all initial conditions, either: i) the solution goes off to ∞ , or ii) The solution is periodic and is either a circle, a straight line, or an ellipse.
- Consider $\ddot{\vec{r}} = -kr^{n-1}\vec{r}$. Pick any n that you like, and any $k > 0$ that you like, but not $n = 1$ nor $n = -2$ (the previous two cases). Show by numerical experiments that a) if you start from rest you get straight line motion towards the origin; and b) if you start with $[x, y, v_x, v_y] = [1, 0, 0, v_y]$ and you pick $\vec{y}$ appropriately you get circular motion.

- (d) For your given k and n ($\neq 1$ or -2) show that for random initial conditions you generally do not get periodic motion.
- (e) For this central force problem, find a periodic motion that is not a straight line and not a circle (using root finding of some kind).
- (f) Using the 2-mass, 2-spring problem we have been toying with in previous weeks, with all damping $= 0$, and with at least one spring having non-zero rest length, with the gravity, and with the anchor points not vertically aligned, find a periodic motion. All else, above, is prep for this problem.

14. **Stability of periodic motion.** For any systems of your choice, can you find:

- a) An unstable periodic motion.
- b) A neutrally stable periodic motion.
- c) An asymptotically stable motion.

You can use a different system for each case, or if you dare, find a single system with two different periodic motions with two different kinds of stability (I don't think you will find a single system with three periodic motions, each one with a different stability case) . In all cases, document your result as clearly and convincingly as you have time for (for example does the eigenvalue analysis agree with the numerical simulation).